### 11.2.1 Hypothalamus

Hypothalamus is a small cone shaped structure that projects downward from the brain ending into the pituitary stalk. It interlinks both the nervous system and endocrine system. Though pituitary gland is known as master endocrine glands that controls the other endocrine glands, but it is, in turn controlled by the hypothalamus. Hypothalamus contains groups of neurosecretory cells. It produces neurotransmitters which regulate the secretions of the pituitary (Figure11. 2). The hormones produced by the hypothalamus act either as a releasing hormone or as an inhibitory hormone.

In the basal region of the brain, the **hypothalamic hypophyseal portal blood vessel** connects hypothalamus and anterior pituitary. It allows hypothalamic hormones to control anterior pituitary secretion. The posterior pituitary is connected with hypothalamus by a nerve bundle called **hypothalamic hypophyseal axis**. It produces nerve signal that control the posterior pituitary secretion. Hypothalamus maintains homeostasis, blood pressure, body temperature, cardio and fluid electrolyte balance of the body. As the part of limbic system it influences various emotional responses.

In mammals, the role of pars intermedia is insignificant, but in other vertebrates it secretes **melanocyte stimulating hormone** (MSH). MSH induces pigmentation in skin.

### 11.2.2 Pituitary gland or Hypophysis

The pituitary gland (means to grow under) is ovoid in shape and is located in the **sella turcica**, a bony cavity of the sphenoid bone at the base of brain and connected to the hypothalamic region of the brain by a stalk called **infundibulum**. It is about one centimetre in diameter and 0.5 gm in weight. The pituitary consists of two lobes, anterior glandular adenohypophysis and posterior neural neurohypophysis. The anterior lobe originates from the embryonic invagination of pharyngeal epithelium called **Rathke's pouch** and the posterior lobe is originates from the base of the brain as an outgrowth of hypothalamus. Anatomically the adenohypophysis has three lobes or zones namely pars intermedia, pars distalis and pars tuberalis. The neurohypophysis is otherwise known as pars nervosa.

**Table 11.2** The major hypothalamic hormones and their functions

| S.No. | Hormones | Functions |
|---|---|---|
| 1. | Thyrotropin releasing hormone (TRH) | Stimulates the secretion of TSH |
| 2. | Gonadotropin releasing hormone(GnRH) | Stimulates the secretion of FSH |
| 3. | Corticotropin releasing hormone (CRH) | Stimulates the secretion of ACTH |
| 4. | Growth hormone releasing hormone(GHRH) | Stimulates the secretion of GH |
| 5. | Prolactin releasing hormone (PRH) | Stimulates the secretion of Prolactin |
| 6. | Luteinizing hormone releasing hormone (LHRH) | Stimulates the secretion of LH |
| 7. | MSH releasing hormone | Stimulates the secretion of MSH |
| 8. | Growth hormone-inhibiting hormone (GHIH) | Inhibits the secretion of GH |
| 9. | Prolactin inhibiting hormone (PIH) | Inhibits the secretion of Prolactin |
| 10. | MSH inhibiting hormone | Inhibits the secretion of MSH |



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